U.S. FOOD & DRUG ADMINISTRATION

Interview Talk with FDA/CDRH

Design for immuno-oncology clinical trials with a heterogeneous patient population

Unkyung Lee

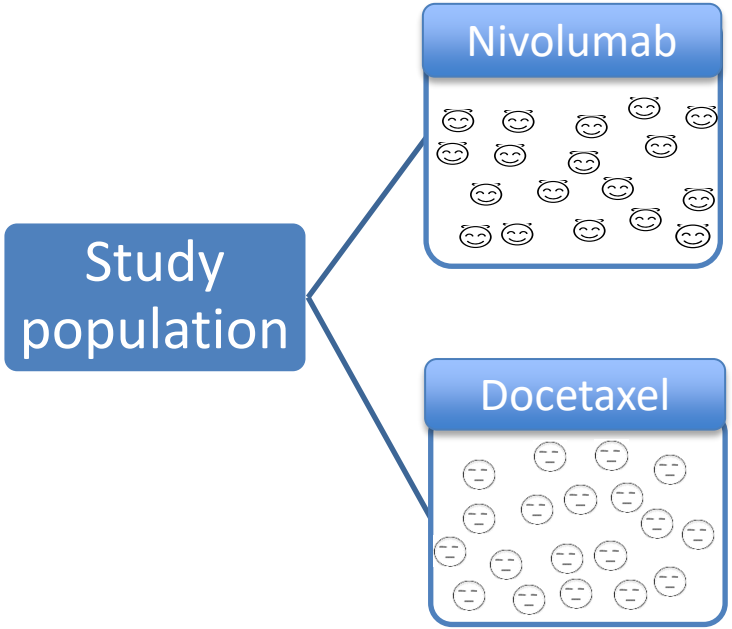
Postdoc ORISE Fellow

Mentor: Dr. Zhenzhen Xu

January, 9th, 2023

Motivating Example

Nivolumab Non-Small-Cell Lung Cancer (NSCLC) Study



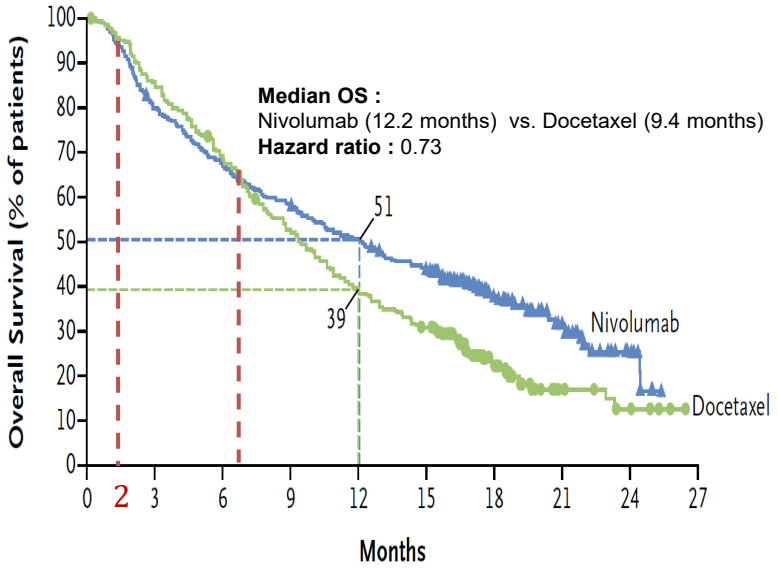
- Primary Objective:
  evaluate efficacy of Nivolumab vs. Docetaxel
- Primary Endpoint : Overall Survival

Nivolumab NSCLC study (Borghaei et al, NEJM, 2015)

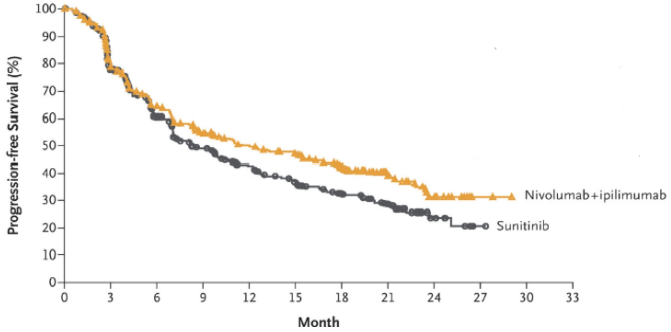
Design Strategy (Borghaei et al, 2015)

- Design: Simulated-based, hybrid design
- Is such design efficient?
  - Cox proportional hazard
  - Log-rank test
- No! Why?
  - Violate proportional hazard assumption
  - The regular log-rank test is not most efficient

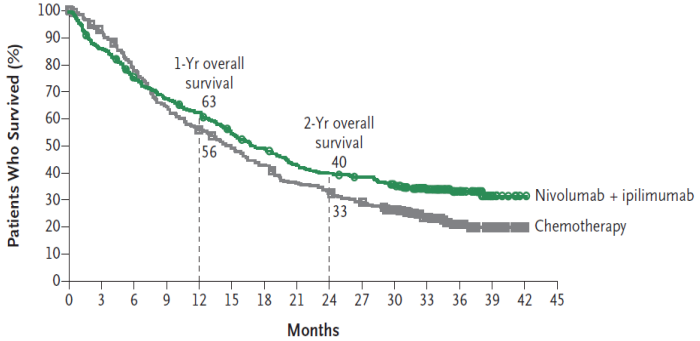
Occurrence of the nonproportional hazard (NPH) patterns



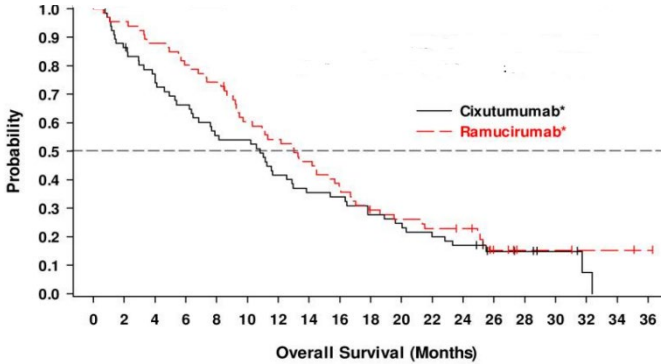
Various NPH patterns of Kaplan–Meier curves in immuno-oncology (IO) trials



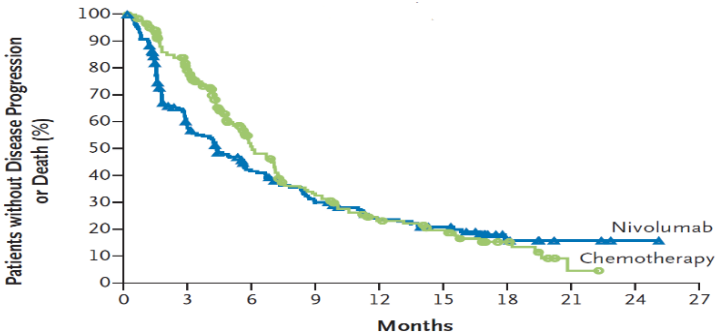
A) Delayed treatment effect pattern



B) Crossing hazard pattern



C) Belly shape diminishing effect pattern

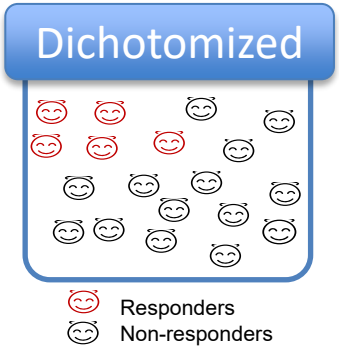


D) Combination pattern

References : (A) Motzer et al., NEJM, 2018 (B) Hellmann et al, NEMJ, 2019 (C) Hussain M, Eur J Cancer 2015 (D) Carbone et al., NEJM, 2017.

PRIME design (Xu et al., 2020)

- PRIME: p%-responders information embedded

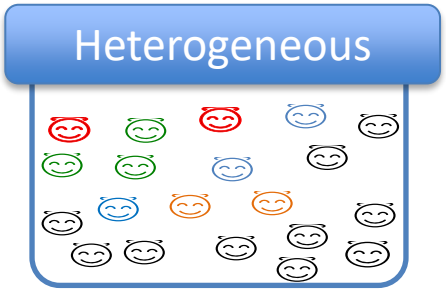


- Assumption: treatment group contains responders vs. nonresponders
- Theory: Dichotomized treatment response would cause the NPH pattern to arise
- Feature:
  - incorporate dichotomized treated response profile in the design and analysis procedure
  - consider delayed effect
- Purpose: Salvage power loss due to NPH issue

Target Power	Design				
90%	PRIME			Simulated-based, hybrid	
	P_{prime}	HR	N	HR	N
	20%	0.01	400	0.73	582

Very Nice, but …

In practice, the study population is heterogeneous rather than dichotomized



Complete response

Partial response

Stable disease

⋮

Nonreponders

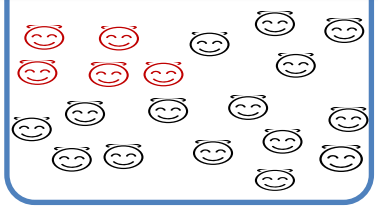
Heterogeneous treatment response profile reported in IO literature:

Cancer Type	Treatment	Reference	Objective Responders (OR)	Patients with Stable disease (SD)	Patients with Progressive disease (PD/NR)	Unknown
NSCLC	Nivolumab	Borghaei et al., NEJM, 2015	19%	25%	44%	11%
Prostate	Cixutumum	Hussain et al., EJC, 2015	15.2%	50%	.	.
Advanced melanoma	ipilimumab	Ribas et al., Jama, 2016	33%	18%	.	.
Advanced Renal-cell carcinoma	Nivolumab plus Ipilimumab	Motzer et al., NEJM, 2018	41%	31%	20%	8%
Melanoma	Nivolumab	Topalian et al., JCO, 2014	31%	7%	58%	4%

Motivation of Extension Work

PRIME

Dichotomized

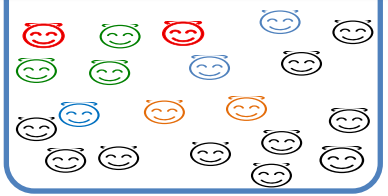


Responders
Non-responders

generalized

Novel Design?

Heterogeneous





Subjects responding to treatment to various extent

Hypothesis

Heterogeneous patient population would be a primary cause of the nonproportional hazard (NPH) patterns.

Analytic Justification

Theorem

The population overall HR, $h(t)$, varies over time except when treatment response is homogeneous or the treatment regimen is not effective at all.

- $$h(t) = \underbrace{\sum_{j=1}^{K-1} h_j}_{\text{Time independent}} - \underbrace{\frac{\sum_{j=1}^{k-1} h_{jc}^{(k-1)} p_j S_c^*(t) + \left\{\left(\sum_{j=1}^{k-1} h_j\right) - 1\right\} p_k S_c^*(t)}{S_T^*(t)}}_{\text{Time dependent}}, \; j = 1,2,\dots,K.$$

  - $h(t)$ become a constant when
    - $h_j = 1 \; for \; all \;\; repsonders \; subgroups \Leftrightarrow \beta_j = 0$ *(No effect for treatment)*
    - $p_j = 1 \; for \; some \; j$ *(Homogeneous treatment group)*

Theorem

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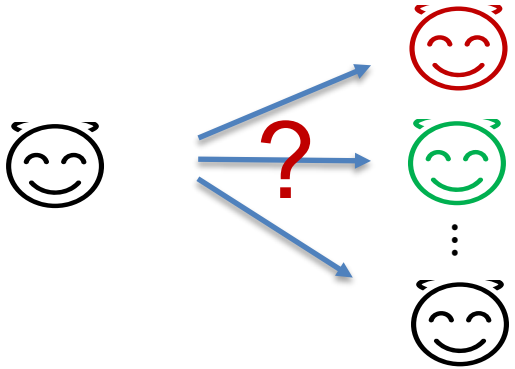
NPH patterns are guaranteed to occur if a heterogeneous patient population is treated by any effective immunotherapy agent.

How to Properly Design with a Heterogeneous Patient Population

Primary cause of NPH pattern:
treating heterogeneous patient population

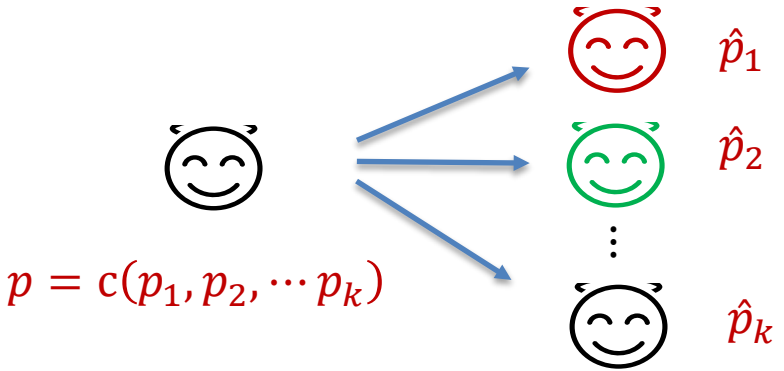
Difficulty:

- **unknown** individual patient's subgroup membership



Solution:

- chance of response with subgroup prevalence $p_j,\ j = 1, 2, \cdots k$
- embeds such information p_j in design and analysis



PRIME+ Strategy

PRIME+

Feature:

- Generalize PRIME to account for more than two responders subgroups
  - Objective responders, stable disease, non-responders
- Incorporate baseline subgroup prevalence to design and analysis procedures
  - $p = \mathrm{c}(p_1, p_2, p_3)$: baseline subgroup prevalence
  - $Z_i \sim Multinomial(N;\ p_1, p_2, p_3)$
- Account for delayed effect

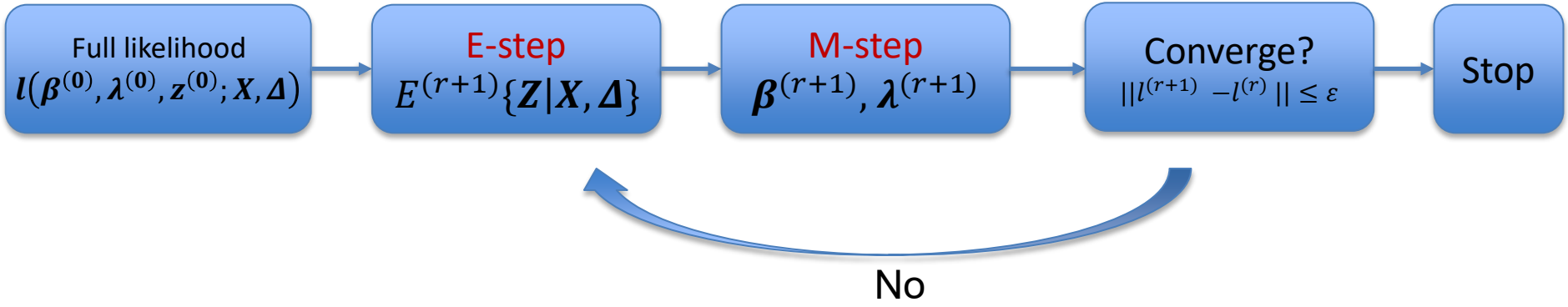
Purpose:

- Improve study efficiency in the presence of NPH patterns
- Provide subgroup-specific treatment effect estimation for $\boldsymbol{\beta} = (\beta_1, \beta_{2,} \beta_3)$

PRIME+

- PRIME+ design and analysis strategy includes
  - Treatment effect estimation: PRIME+ EM algorithm
  - Hypothesis testing: PRIME+ likelihood ratio test and Re-randomization test
  - Study design: PRIME+ sample size and power calculation
- R package (Immunotherapy.Design) is developed for implementation.

PRIME+ EM Algorithm



PRIME+ Testing Procedures

- Hypothesis: $H_0: \beta_1 = \beta_2 = \beta_3 = 0, \quad vs. \quad H_a: \ at \ least \ one \ \beta_j \neq 0, \ j = 1, 2, 3$

- Parametric Testing procedure: PRIME+ likelihood ratio test
  - $-2\{\log \Lambda(x, \Delta)\} = 2\{l_m(\hat{\beta}^{(k)}, \hat{\lambda}^{(k)}; X, \Delta) - l_m(\hat{\beta}^0, \hat{\lambda}^0; X, \Delta)\} \sim \chi_2^2$

- Nonparametric Testing procedure: PRIME+ re-randomization test
  - Create an empirical reference distribution of observed test statistic

PRIME+ Sample Size and Power Calculation

Given target power, sample size searching algorithm based on bisection method

Step 1. Set initial sample size searching interval $[N_{min,0}, N_{max,0}]$ such that $EP_{N_{min,0}}$ < target power < $EP_{N_{max,0}}$

Step 2. Update a midpoint : $N_{mid,i} = \frac{N_{max,i}+N_{min,i}}{2}$ and estimate EP at $N_{mid,i}$

Test if **empirical power at $N_{mid,i}$ is closest to target power**

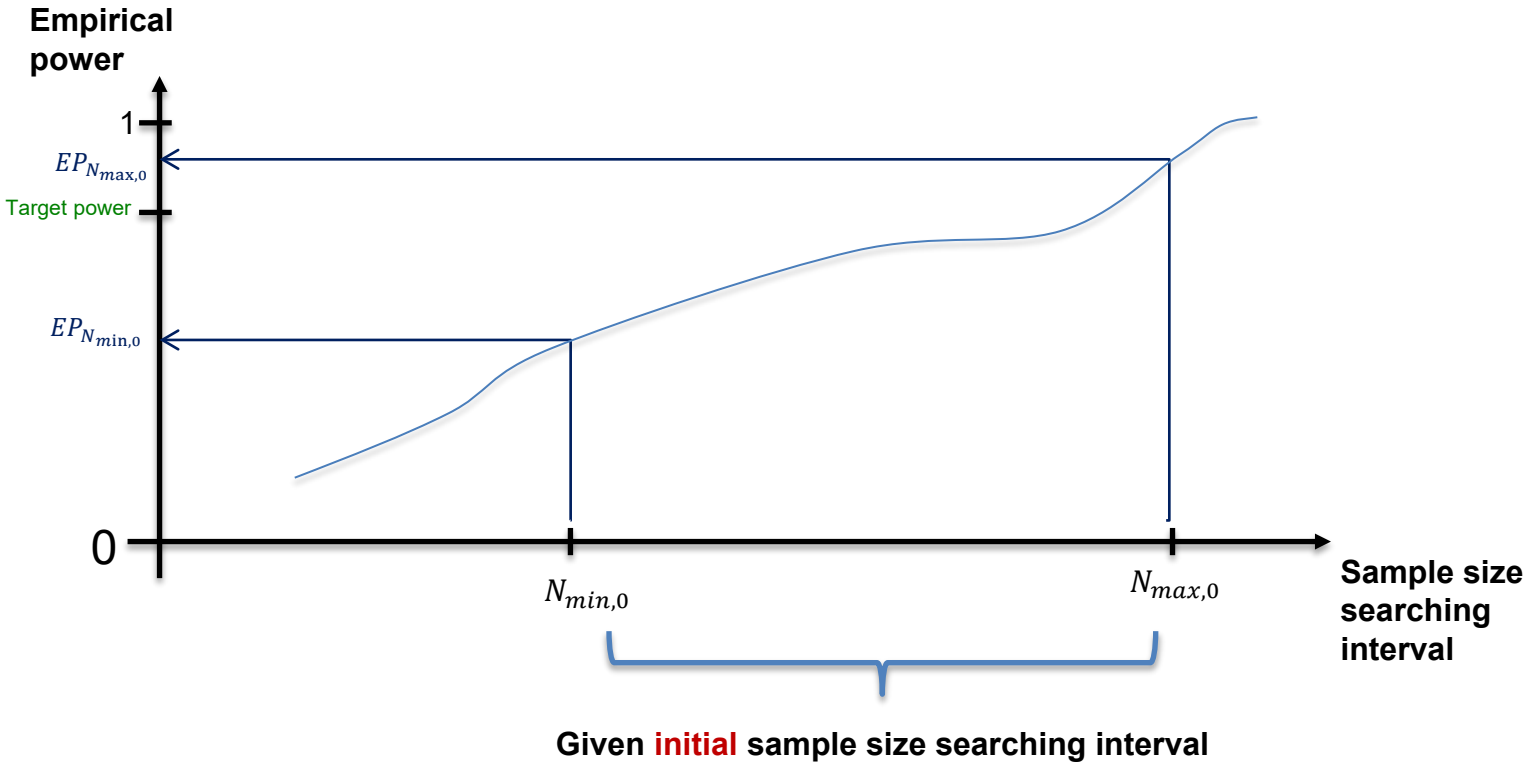
Yes → **Stop** Return $N_{mid,i}$

No

Step 3. Update the search interval to be $[N_{min,i}, N_{max,i}= N_{mid,i}]$ Or $[N_{min,i} = N_{mid,i}, N_{max,i}]$

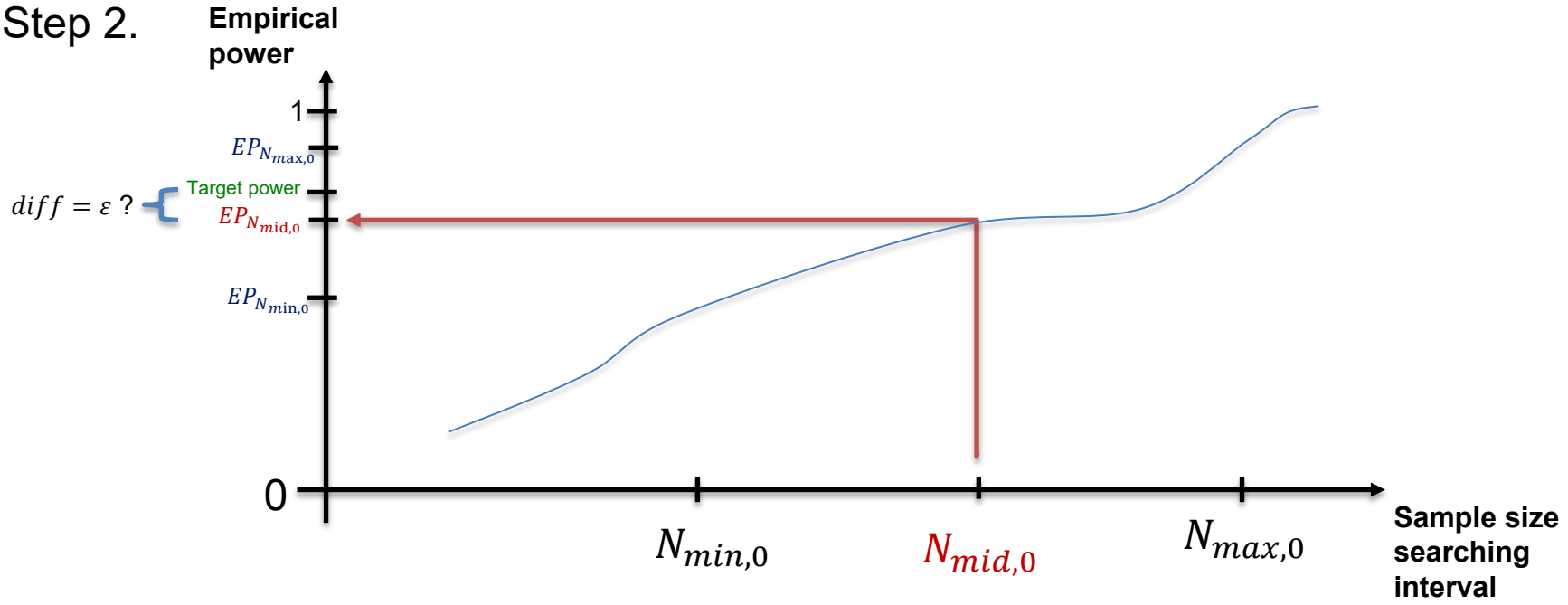
PRIME+ Sample Size Searching Algorithm

Step 1.



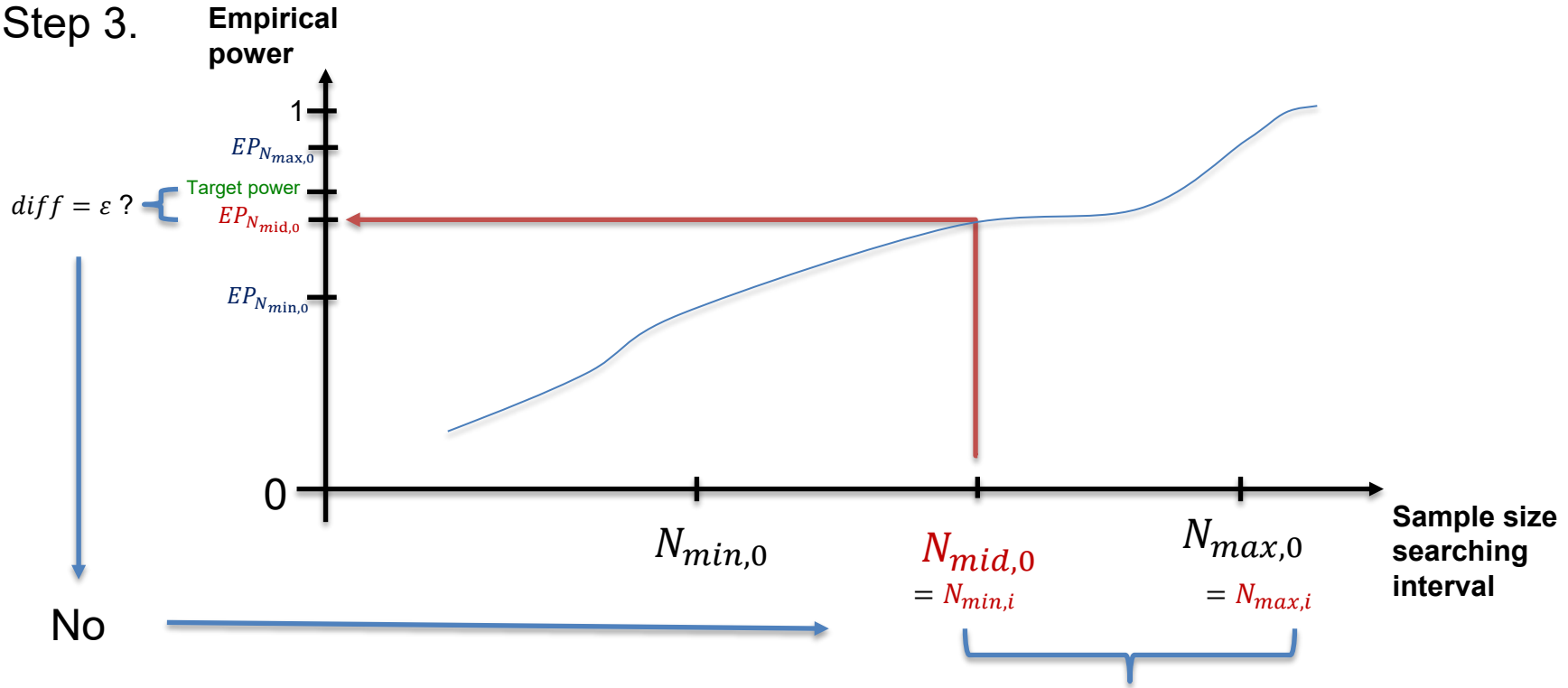
PRIME+ Sample Size Searching Algorithm

Step 2.



PRIME+ Sample Size Searching Algorithm

Step 3.



Simulation Study

Evaluation #1. Properties of Treatment Effects Estimation

		Objective Responders			Subjects with stable disease		
P	β	Bias$(\hat{\beta}_1)$	ESE$(\hat{\beta}_1)$	CP$(\hat{\beta}_1)$	Bias$(\hat{\beta}_2)$	ESE$(\hat{\beta}_2)$	CP $(\hat{\beta}_2)$
(0.35, 0.35, 0.30)	(log(0.2), log(0.4), 0)	0.004	0.482	0.957	0.000	0.459	0.966
	(log(0.3), log(0.6), 0)	0.039	0.368	0.963	0.005	0.439	0.969
	(0, 0, 0)	0.015	0.187	0.953	0.015	0.187	0.953

Lag duration is 1 month. Baseline hazard is 0.117. Total study duration is 60 months, and enrollment rate is 15.8 subjects per month. Sample size is 300. The estimates are obtained from 1000 simulation runs.

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Evaluation #2. Power Comparison

Design		Analysis		
		PRIME +		Conventional
		LR Test	Re-random test	Log –rank test
P	β	EP	EP	EP
(0.20, 0.30, 0.50)	(0, 0, 0)	0.019	0.033	0.036
	(log(0.4), log(0.7), 0)	0.625	0.702	0.569
	(log(0.3), log(0.6), 0)	0.893	0.942	0.820
	(log(0.2), log(0.5), 0)	0.977	0.998	0.966

5% of Type 1 error rate and baseline hazard is $\log(2)/3.5$. Lag duration is 2 months. Total study duration is 60 months. Enrollment rate is 400/15.8. Sample size is 400. EP: empirical powers

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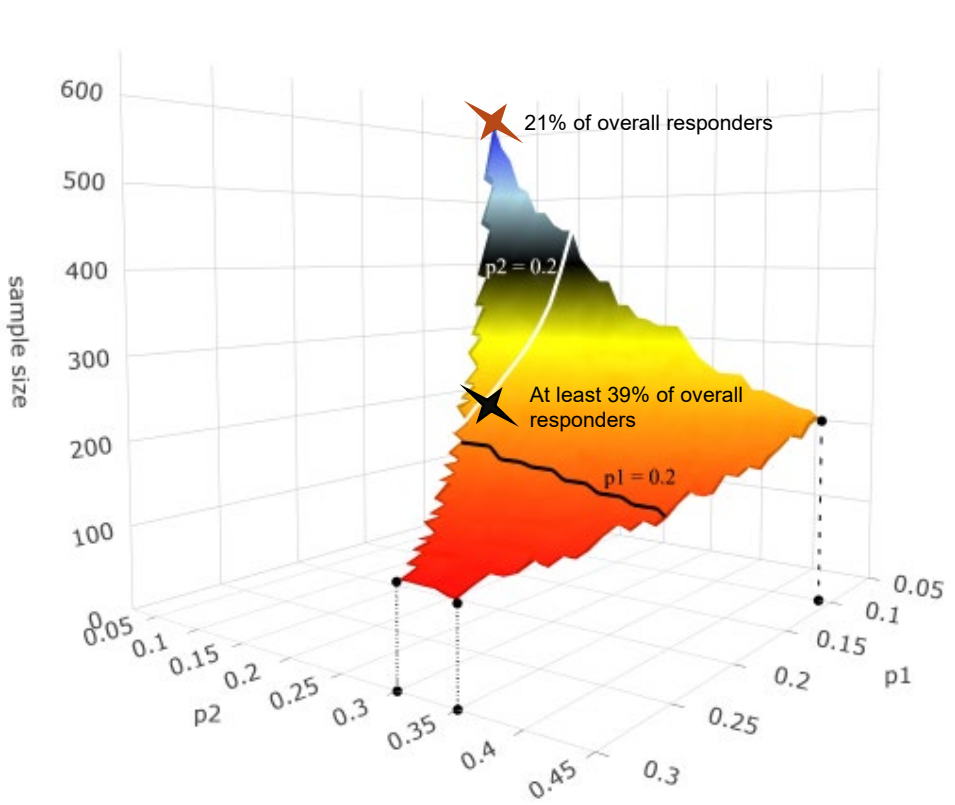
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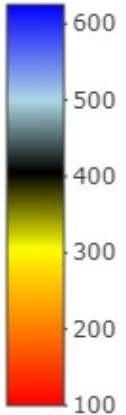
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Evaluation #3.1. Impact of treatment response heterogeneity on study efficiency



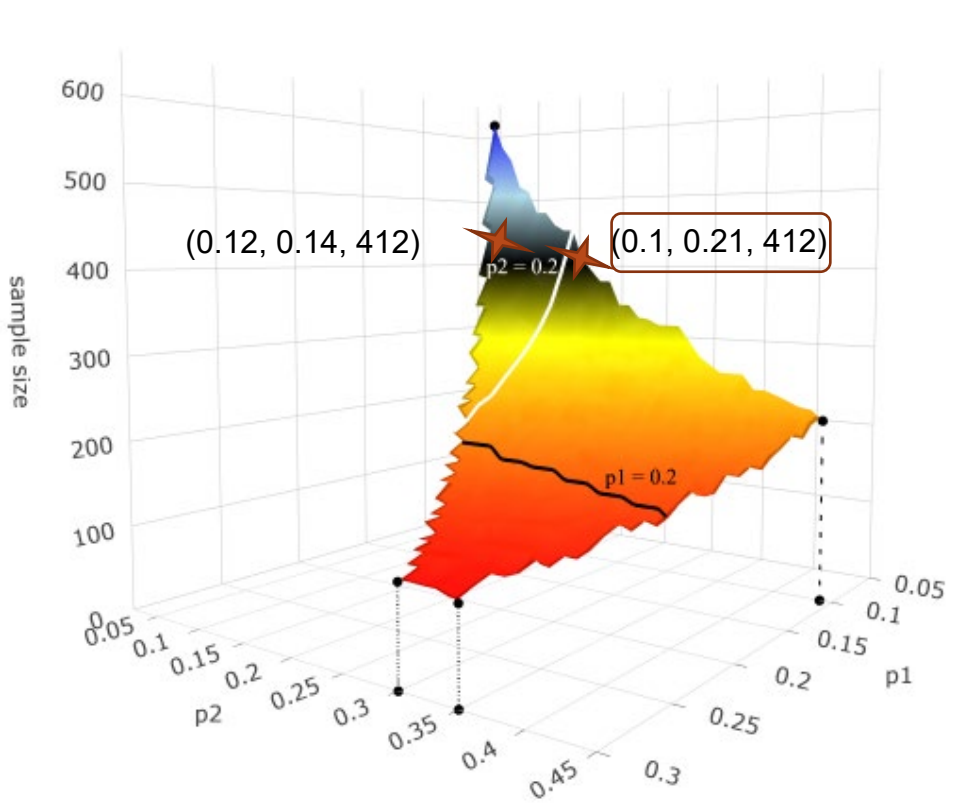
Sample size



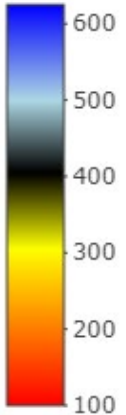
p_1 : Prevalence of objective responders
p_2 : Prevalence of subjects with stable disease

80% of target power
HR=c(0.2, 0.52, 1)

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Sample size



p_1 : Prevalence of objective responders
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Evaluation #3.2. Impact of treatment response heterogeneity on study efficiency

Design								
PRIME+				PRIME			Conventional	
		LRT	Re-random					
P	β	N	N	P_{prime}	$\hat{\beta}_{prime}$	N	$\hat{\beta}_{conv}$	N
(0.15, 0.25, 0.60)		529	432	0.40	log(0.38)	359	log(0.75)	523
(0.20, 0.25, 0.55)	(log(0.2), log(0.52), 0)	337	298	0.45	log(0.36)	262	log(0.72)	404
(0.30, 0.35, 0.35)		155	132	0.65	log(0.35)	118	log(0.60)	177

Target power is 80%. Lag duration is 2 months. Total study duration is 29months. Baseline hazard is 0.0737. Enrollment rate is 36.8 subjects per month. The type 1 error rate is 0.05.

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(0.30, 0.35, 0.35)		155	132	0.65	log(0.35)	118	log(0.60)	177

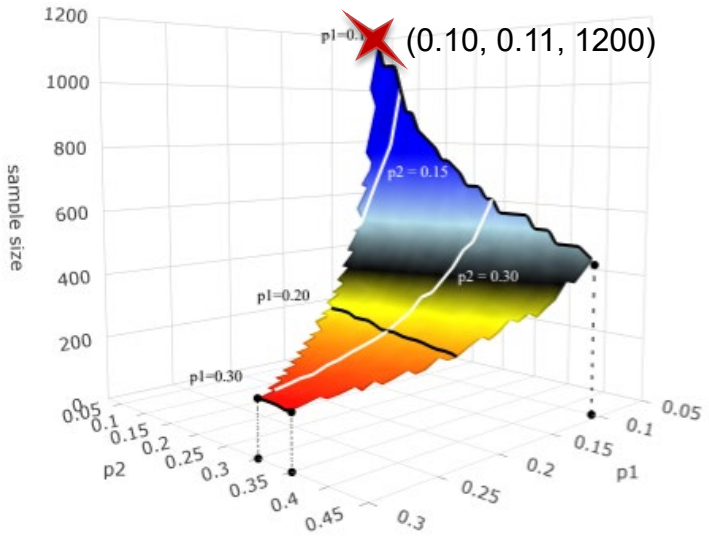
Target power is 80%. Lag duration is 2 months. Total study duration is 29months. Baseline hazard is 0.0737. Enrollement rate is 36.8 subjects per month. The type 1 error rate is 0.05.

Evaluation #3.2. Impact of treatment response heterogeneity on study efficiency

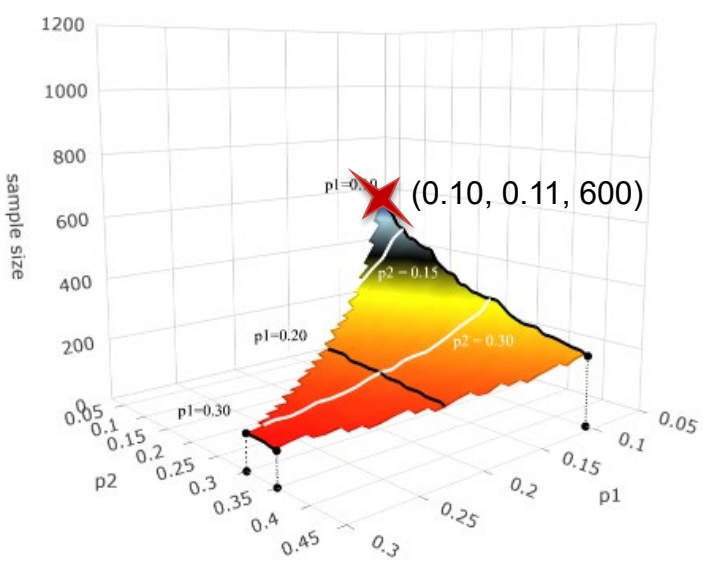
Design								
PRIME+				PRIME			Conventional	
		LRT	Re-random					
P	β	N	N	P_{prime}	$\hat{\beta}_{prime}$	N	$\hat{\beta}_{conv}$	N
(0.15, 0.25, 0.60)	(log(0.2), log(0.52), 0)	529	432	0.40	log(0.38)	359	log(0.75)	523
(0.20, 0.25, 0.55)		337	298	0.45	log(0.36)	262	log(0.72)	404
(0.30, 0.35, 0.35)		155	132	0.65	log(0.35)	118	log(0.60)	177

Target power is 80%. Lag duration is 2 months. Total study duration is 29months. Baseline hazard is 0.0737. Enrollement rate is 36.8 subjects per month. The type 1 error rate is 0.05.

Evaluation #4. Impact of effect size on the relationship between treatment response heterogeneity and study efficiency

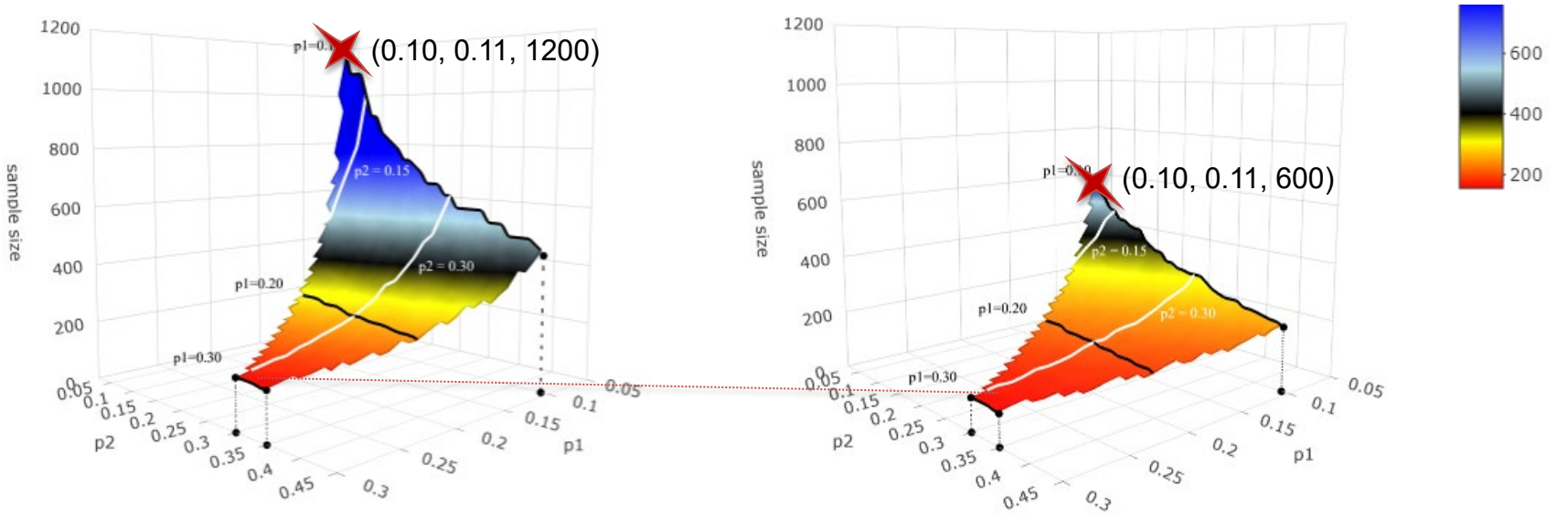


(a) HR=c(0.25, 0.70, 1)



(b) HR=c(0.20, 0.52, 1)

Evaluation #4. Impact of effect size on the relationship between treatment response heterogeneity and study efficiency



(a) HR=c(0.25, 0.70, 1)

(b) HR=c(0.20, 0.52, 1)

Redesign Motivating Example

Redesign Nivolumab Study using PRIME+ strategy

- PRIME+ strategy reflects the features of nivolumab study (Borghaei and others, 2015)
  - Observed lag time: 2.1 months
  - Treatment response heterogeneity: 19% of OR, 25% of SD, and 44% of NR
  - Observed treatment effect: 0.73 for overall HR between nivolumab vs docetaxel

- Redesign the study using PRIME+ :
  - Prevalence of each responder subgroup
    - $p_1 = 0.20$
    - $p_2 = 0.25$
    - $p_3 = 0.55$
  - Subgroup specific treatment effects
    - HR for OR : 0.2
    - HR for SD : 0.52
    - HR for NR : 1

} 0.73 for average HR for nivolumab group

Redesign Nivolumab Study using PRIME+ strategy

Target Power	Redesign			Original Design	
90%	PRIME+			Hybrid- simulated	
	p	HR	N	HR	N
	(0.2, 0.25, 0.55)	(0.20, 0.52, 1)	450	0.73	582

Redesign Nivolumab Study using PRIME+ strategy

Target Power	Redesign			Original Design	
90%	PRIME+			Hybrid- simulated	
	p	HR	N	HR	N
	(0.2, 0.25, 0.55)	(0.20, 0.52, 1)	450	0.73	582

Conclusion

- PRIME+ improves PRIME when patient population is heterogeneous
- Detect a subgroup specific treatment effect profile
  - Offer insight to investigators to properly administer treatment
  - Report comprehensive information to prescribers
- Salvage power loss due to violation of PH assumption
  - by properly accounting for treatment response heterogeneity
- Outline a path to improve study efficiency
  - by treating more responsive subjects

Thank You!

Selected References

- Borghaei H, Paz-Ares L, Horn L, et al. **Nivolumab versus docetaxel in advanced nonsquamous non-small-cell lung cancer**. *N Engl J Med., 2015.*

- Xu Z, Zhu B, Park Y. **Design for immuno-oncology clinical trials enrolling both responders and nonresponders**, *Statistics in Medicine, 2020.*

- Dempster AP, Laird NM, Rubin DB. **Maximum likelihood from incomplete data via the em algorithm**, *J Royal Stat Soc Ser B, 1977.*

Questions and Comments



Back Up Slides

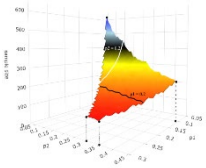
PRIME+ Power Calculation

Given sample size N, we estimate power for responder specific treatment effects $\beta = (\beta_1, \beta_2, \beta_3)$ by following the below procedure,

- Step 1. Prespecify all necessary parameters including $p = (p_1, p_2, p_3)$.
- Step 2. Simulate survival outcomes using prespecified p and event time;
  - $T_j \sim pexp(h_c, h_E)$ for each responder subgrouop, where rate changes at t^*
  - T_k *or* $T_c \sim exp(h_c)$ for nonresponders and control group,

  where h_c is baseline hazard and h_E hazard for each responders subgroup after t^*.
- Step 3. Apply PRIME+ likelihood ratio test or PRIME+ re-randomization test to compute p-value.
- Step 4. Repeat steps 2 and 3 for 1000 times.
- Step 5. Compute the empirical power for given marginal likelihood l_m
  as the proportion of iterations whose estimated p-values are less than equal to $\alpha = 0.05$.

Evaluation #3.1. Impact of treatment response heterogeneity on study efficiency



2D

p_1 : Prevalence of objective responders
p_2 : Prevalence of subjects with stable disease

Constraints : $p_1 \leq p_2 \leq p_3,\ \sum_{j=1}^{3} p_j$

31% Overall Responders

26% Overall responders

p2: 0.45, 0.40, 0.35, 0.30, 0.25, 0.20, 0.15, 0.10

p1: 0.10, 0.12, 0.14, 0.16, 0.18, 0.20, 0.22, 0.24, 0.26, 0.28, 0.30

size: 600, 500, 400, 300, 200, 100

Evaluation #3.2. How To Prespecify Underlying Subgroup Specific HRs

Step 1. Choose a plausible treatment effect size for each responder subgroup $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3)$.
Step 2. Prespecified $\boldsymbol{p} = \mathrm{c}(p_1, p_2, p_3)$.
Step 3. Given $\boldsymbol{\beta}$ and $\boldsymbol{p}$, simulate survival outcomes;

- $T_j \sim pexp(h_c, h_E)$ for each responder subgroup, where rate changes at t^*
- $T_k \; or \; T_c \sim exp(h_c)$ for nonresponders and control group,

where h_c is baseline hazard and h_E hazard for each subgroup after t^*

Step 4. Apply Cox PH model to estimate the averaged HR between treatment group and control group $\beta_{conv}^{(b)}$.
Step 5. Repeat steps 3-4 for $b = 1000$ times and record $\beta_{conv}^{(1)}, \beta_{conv}^{(2)} \ldots, \beta_{conv}^{(1000)}$.
Step 6. Average out the estimates of $\beta_{conv}^{(b)}$ across 1000 times and record $\bar{\beta}_{conv}$.

- Vary a plausible treatment effect sizes for $\boldsymbol{\beta} = (\beta_1, \beta_2, \beta_3)$.
- Repeat above procedure until $\bar{\beta}_{conv}$ is closest to log(0.73) for average HR for treatment group from Nivolumab study.

Evaluation #3.2. How To Estimate Responder Specific HR under PRIME

Step 1. Given prespecified $\boldsymbol{p} = \mathrm{c}(p_1, p_2, p_3)$ and $\boldsymbol{\beta} = (\beta_1, \beta_{2,} \beta_3)$**,** simulate survival outcomes.

Step 2. Combine two responders subgroups to a overall responder subgroup and estimate p_{prime}.

Step 2. Apply PRIME EM algorithm to estimate the responder specific HR $\beta_{prime}^{(b)}$.

Step 3. Repeat steps 1-3 for $b = 1000$ times and record $\beta_{prime}^{(1)}, \beta_{prime}^{(2)} \ldots, \beta_{prime}^{(1000)}$.

Step 4. Average out the estimates of $\beta_{prime}^{(b)}$ across 1000 times and record $\bar{\beta}_{prime}$.

Evaluation #3.2. How To Estimate Average HR Under Conventional

Step 1. Given prespecified $\boldsymbol{p} = \mathrm{c}(p_1, p_2, p_3)$ and $\boldsymbol{\beta} = (\beta_1, \beta_{2,} \beta_3)$**,** simulate survival outcomes.
Step 2. Combine all responders subgroups and nonresponders for a treatment group.
Step 2. Apply Cox proportional hazard model to estimate the averaged HR $\beta_{conv}^{(b)}$.
Step 3. Repeat steps 1-3 for $b = 1000$ times and record $\beta_{conv}^{(1)}, \beta_{conv}^{(2)} \dots, \beta_{conv}^{(1000)}$.
Step 4. Average out the estimates of $\beta_{conv}^{(b)}$ across 1000 times and record $\bar{\beta}_{conv}$.

Redesign: Searching Algorithm For Subgroup Specific HR Under PRIME+

Given a plausible subgroup specific HR for OR, SD and PD/NR: $\beta = (\beta_1, \beta_2, \beta_3)$,

Step 1. Simulate survival outcomes under 1:1 treatment allocation ratio.
Step 2. Apply Cox PH model to estimate an overall treatment effect $\beta_{conv,i}$.
Step 3. Repeat steps 1 and 2 for 1000 times.
Step 4. Record $\beta_{conv,i}$ for $i = 1,2,\cdots,1000$.

- $\bar{\beta}_{conv}$: Average out those estimated overall treatment effects across 100 simulations.
- Repeat the above entire algorithm and find β until the estimated $\bar{\beta}_{conv}$ is closest to the observed HR 0.73.